ARCHITECTURAL SPECIFICATION

Amagi-MPS v2.1

Master Parameter Sheet — Hardware-Enforced PTCB for Autonomous CPS

VERSION 2.1 — OCTOBER 2026

> Canonical parameter reference for the AMAGI Architectural Class. Structured in two layers: **Layer A** — substrate-independent architectural parameters (the canon of the class); **Layer I** — parameters of concrete implementation instances (RI-1, RI-2), recorded as dated evidence, not as canon. Safety-integrity figures are normative **only in FMEDA v1.1**; this sheet cites them. Aligned with: IAA v1.1, Charter v1.2, S-APL v2.0, Framework v2.0 (corrigendum v2.1 Planned), NIPU v1.1, MEM v1.1, Open Specification v1.1, NIST AI RMF Profile v1.1, HARA v1.1, FMEDA v1.1, Traceability Matrix v1.1, Master Glossary v1.0.

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1. Purpose

This document consolidates the quantitative parameters of the AMAGI ecosystem into a single authoritative source. Version 2.1 supersedes v2.0 and resolves the parameter conflicts identified during the 2026 consistency freeze by introducing an explicit two-layer structure and by delegating quantitative reliability figures to the FMEDA as the single normative source.

1.1 Layer Model (normative for this sheet)

Layer	Content	Change policy
A — Architecture (canon)	Invariants INV-001…006, SCC protocol semantics, TTP predicate, MAG target structure, crypto migration path, isolation profiles S/D, integrity capability formula (§3.1)	Changes only via deliberate major revision, cross-referenced in all aligned documents
I — Implementation instances	Measured/verified parameters of concrete builds: **RI-1** (VHDL prototype, 2026-05) and **RI-2** (SystemVerilog reference implementation, per AMAGI TCAD manuscript TCAD-2026-1011)	Append-only register; instances are never merged or averaged. New instances are added, existing entries are frozen

A parameter without a layer tag is void. Where earlier ecosystem documents state a bare number (e.g., “SCC VHDL lines ~340”, “states explored 2.43E6”), it shall be read as an RI-1 attribute, not as a class-canonical value.

1.2 Implementation Instance Register (summary)

ID	HDL	Lines	Role	Primary evidence
RI-1	VHDL	~340	Historical prototype (benchtop measurements)	AF v1.4; MPS v2.0; SPIN v6.5 run
RI-2	SystemVerilog	467	**Current reference implementation**	TCAD-2026-1011 (rejected 2026-09-22; new submission in preparation; Canonical RI per VC v1.1 §6); NIPU v1.1; independent SPIN re-run during TCAD peer review (Reviewer 2, statistics confirmed)

1.3 Isolation Profiles for INV-001 (D5)

Profile	Scope	Status
S — Subsystem isolation (reference)	INV-001 enforced at PTCB subsystem boundary; host AI stack shares the device	Current, RI-1/RI-2
D — Device-level isolation (target)	PTCB physically separated at device level; AI host cannot share die/board memory path with the enforcement core	Phase 2 (2027); closes device-level-isolation review finding

1.4 Important Notice

This document contains architectural parameters only. Implementation-level details remain protected under the AMAGI Framework Trade Secrets Notice (DOI: 10.5281/zenodo.17886699) and the S-APL.

2. Temporal Parameters (Temporal Triad)

> **Terminology note (C7):** «Temporal Triad» denotes the temporal structure Trigger → Programmed Delay → Localized Activation. The acronym **TTP** in AMAGI denotes the **Tautological Truth Predicate** — the formal predicate enforced by the SCC (see FMEDA v1.1, Master Glossary v1.0). The v2.0 reading «Temporal Triad Parameters (TTP)» is withdrawn.

All temporal parameters are deterministic design parameters; the verification status of each value is stated per instance in the Evidence column (rule C8). The parameter table is unchanged from v2.0 (§2, including the Latency Decomposition Formula), with the following layer annotations:

- t_cutoff = 18–42 µs — **RI-1**, full-implementation envelope, benchtop measurement;
- t_cutoff = 12–35 µs — **TCB-only envelope** per Framework v2.0 / NIPU v1.1 (class-level specified bound; RI-2 board-level measurement not yet performed — Phase 2).

Domain inter-phase delays (§2, t_med / t_auto / t_aero) — Layer A (domain requirements imported from external standards).

Parameter	Symbol	Measured	Specified	Theoretical	Unit	Source	Verification
Policy verification latency	t_policy	—	1-10	0.001-1000	us	AF-COMP v1.4	Design target based on SCC architecture
Comparator delay	t_comp	~0.2	<=0.2	>=0.001	us	Benchtop prototype	Measured; typical for modern fast comparators
FPGA sync delay	t_sync	0-0.01	<=0.01	>=0.001	us	Benchtop prototype	Single-cycle sync; verified by SPIN model
FSM logic delay	t_fsm	~0.2	<=0.2	>=0.001	us	Benchtop prototype	Modelled in SPIN; consistent with synthesized HW
MOSFET switching	t_mosfet	10-40	10-40	5-100	us	Benchtop prototype	Standard power MOSFET switching times
Total cutoff latency	t_cutoff	18-42	<50	5.2-1040	us	Benchtop prototype	Summed chain; verified by end-to-end measurement
ATL delay range	t_atl	—	—	0.5-1000	ns	MEM-SPEC v1.0	Architectural target
TDF delay range	t_tdf	—	—	1-1000	ns	NIPU-SPEC v1.0	Architectural target

Parameter	Symbol	Measured	Specified	Theoretical	Unit	Source	Verification
Watchdog timeout	t_wdg	30	OTP-conf.	0.001-3600	s	AF-COMP v1.4	Configurable; 30 s default per IEC 61508-2 Table A.13
Inter-phase delay (medical)	t_med	—	>=200	—	ms	IEC 60601-2-24	Requirement confirmed in IEC 60601-2-24, Clause 201.12
Inter-phase delay (automotive)	t_auto	—	>=200	—	ms	ISO 26262 ASIL-D	Consistent with ASIL-D timing constraints
Inter-phase delay (aerospace)	t_aero	—	>=500	—	ms	EASA CS-25	Verified from EASA CS-25 Appendix K
Isolation barrier delay	t_iso	~0.2	<=0.5	0.01-10	us	Benchtop est.	Typical digital isolator (e.g., ISO78xx)
Actuator command latency	t_act	—	<=5	0.1-100	us	AF actuator model	Within FPGA logic + driver delay
Sensor sample-to-decision	t_sens	—	<=50	1-1000	us	CPS design target	Feasible with fast ADC + SCC pipeline
Phase transition completion	t_trans	—	<=100	10-500	us	AF-COMP phase logic	Upper bound for phase handover
Inter-channel skew	t_skew	—	<=0.01	0-0.1	ns	FPGA clock network	Within clock distribution budget
Clock jitter (cycle-to-cycle)	t_jitter	<0.05	<=0.1	>=0.001	ns	Clock gen spec (RMS)	Typical crystal+FPGA PLL jitter

Latency Decomposition Formula

t_cutoff = t_comp + t_sync + t_fsm + t_mosfet + t_margin

Where each component represents: t_comp = analogue comparator (~0.2 us), t_sync = FPGA synchronizer (0-0.01 us), t_fsm = SCC logic (~0.2 us), t_mosfet = power stage switching (10-40 us), t_margin = safety margin (varies by domain).

3. Safety Integrity Parameters

3.1 Integrity Capability Formula (Layer A — canon, D2)

> **SIL 2 @ 1oo1 (HFT=0) — reference configuration, current. SIL 3 @ 1oo2 (HFT=1, β=0.10) — enhanced configuration, Phase 2 target (2027).**
>
> In both configurations the **binding constraint is the architectural constraint** per IEC 61508-2:2010 Table 2 (Type B, SFF): 1oo1 caps at SIL 2, 1oo2 reaches SIL 3. The computed PFDavg values exceed the numeric requirement of IEC 61508-1:2010 Table 3 by at least one decade (see FMEDA v1.1 §7–8); probabilistic margin is therefore not the limiting factor. HARA v1.1 maps hazards to these two deployment configurations.

3.2 Quantitative Parameters (cited, Layer I-agnostic)

Quantitative reliability parameters are **normative exclusively in FMEDA v1.1**. This sheet cites and does not independently recompute:

Parameter	Value	Configuration	Source
PFDavg	6.9×10^{-5}	1oo1	FMEDA v1.1 §7.1
PFDavg	6.9×10^{-6}	1oo2 (β=0.10)	FMEDA v1.1 §7.2

Parameter	Value	Configuration	Source
SFF	91.6% (component) / 93% (system)	—	FMEDA v1.1 §5
Subsystem type	**Type B** (IEC 61508-2)	—	FMEDA v1.1 (corrects v2.0 “Type A”)
Proof test interval **T1**	**8 760 h (1 year)** — normative in all profiles	—	FMEDA v1.1 §7.1
Mission / design life **T_M**	**87 600 h (10 years)**	—	AF v2.0 §lifecycle; see 3.3
CCF β	0.10 (moderate defences); enhanced target 0.02	1oo2	FMEDA v1.1; IEC 61508-6 Annex D

The following v2.0 figures are **withdrawn** as superseded and internally inconsistent with the FMEDA: PFD1 3.6E-4, PFD1oo1 3.6E-4, PFD2 1.8E-4, PFD2p 3.6E-4, PFD1oo2_02 9.0E-5, SFF 92%, “Subsystem type A”, SIL table “Current 2 / Target 3” without configuration binding. Any downstream document still carrying them is non-conformant and shall be corrected (HARA v1.1, TM v1.1 already aligned).

3.3 T1 vs T_M — terminology resolution (C4, D3’)

Earlier corpus documents state both “TI = 1 year” (FMEDA) and “10 years” (Framework v2.0 lifecycle). These are **not contradictory**: 1 year is the **proof test interval T1** (enters PFDavg directly), 10 years is the **mission/design life T_M** (87,600 h). Normative rule: T1 = 8 760 h in every deployment profile; T_M = 87 600 h. Framework v2.0 carries a corrigendum (v2.1, Planned) relabelling its “10-year TI” wording as “10-year design life”. (Resolves fork D3 without introducing two mission profiles — the contradiction was terminological.)

3.4 ASIL mapping (domain role only, C9)

ISO 26262 / ASIL figures are retained as **domain deployment mapping** (§7.2, §15 roadmap), not as the normative integrity frame. ASIL-B baseline / ASIL-D target per ISO 26262-9 decomposition; full ISO 26262 route is a Phase 2 certification activity. The normative frame of this sheet is IEC 61508.

4. Cryptographic Parameters

§4 and the Cryptography Migration Path are unchanged from v2.0, with the FIPS 203/204 renames per Master Glossary D-13 (**Kyber → ML-KEM-1024**, **Dilithium → ML-DSA-44**; parameter sets unchanged) applied throughout §4 and §19. **§4.1 edit (C11):** the SPHINCS+ row is replaced:

Algorithm	Parameter	Current	Target	Unit	Source	Evidence
SLH-DSA-SHA2-128f	Signature	—	7 856	bytes	NIST FIPS 205	AF v2.0 §(resources); auxiliary hash per NIPU v1.1, MEM v1.1

Cryptographic parameters define the security mechanisms used for authentication, key management, and data protection within the Amagi Framework. The architecture supports both current NIST-approved algorithms and post-quantum cryptography (PQC) migration.

Parameter	Symbol	Current	Target	Theoretical	Unit	Source	Verification
Primary algorithm	—	HMAC-SHA256	ML-KEM-1024 + ML-DSA-44	PQC hybrid	—	NIST FIPS 205/203	HMAC-SHA256 per FIPS 198-1; ML-KEM-1024 + ML-DSA-44 per FIPS 203/204
Key generation	—	SRAM-PUF	SRAM-PUF	Quantum RNG	—	NIST SP 800-90B	PUF entropy validated per NIST SP 800-90B

Parameter	Symbol	Current	Target	Theoretical	Unit	Source	Verification
PUF min-entropy	H_min	>=256	>=256	1024	bits	NIST SP 800-90B	>=256 bits confirmed by statistical tests
Hash chain	—	HMAC-SHA256	HMAC-SHA256	SHA-3	—	NIST FIPS 202	SHA-3 defined in FIPS 202
Post-quantum upgrade	—	Upgrade path	Native	Native	—	NIST PQC Standardization	Hybrid mode compatible
Signature scheme	—	HMAC-based	ML-DSA-44	SLH-DSA-SHA2-128f	—	NIST FIPS 205	ML-DSA-44 (FIPS 204), SLH-DSA-SHA2-128f (FIPS 205)
Encryption (memory)	—	—	AES-256 + ML-KEM-1024	AES-256 + ML-KEM-1024	—	MEM-SPEC v1.0	Target AES-256-GCM per NIST SP 800-38D
Key binding	—	HW root	HW root	Quantum-res.	—	NIPU-SPEC v1.0	Hardware root of trust

Cryptography Migration Path

Phase 1 (Current): HMAC-SHA256 + SRAM-PUF

Phase 2 (Target): ML-KEM-1024 (KEM) + ML-DSA-44 (signatures) + SRAM-PUF

Phase 3 (Future): Full PQC hybrid + quantum-resistant key generation

4.1 Cryptographic Performance and Key Sizes — Verified

Algorithm	Parameter	Current	Target	Theoretical	Unit	Source	Verification
HMAC-SHA256	Key size	256	256	256	bits	NIST FIPS 198-1	Key size verified per FIPS 198-1, SP 800-107
	HMAC tag size	256	256	256	bits	NIST	Output tag = 256 bits
	Throughput	1200	1500	5000	Mbps	Meas./ Est.	Measured on FPGA prototype
	Latency (tag)	1.2	<1	0.5	us	Meas./ Est.	Timing from HMAC pipe
ML-KEM-1024 (FIPS 203)	Public key	—	1568	1568	bytes	NIST FIPS 203	Verified: exact value per FIPS 203
	Private key	—	3168	3168	bytes	NIST FIPS 203	Verified: per FIPS 203
	Ciphertext	—	1568	1568	bytes	NIST FIPS 203	Verified
	Key gen time	—	0.1	0.05	ms	Target impl.	Typical for HW accelerator
	Encaps speed	—	3000	5000	ops/s	Target impl.	Achievable in FPGA
	Decaps speed	—	5000	7000	ops/s	Target impl.	Achievable
ML-DSA-44 (FIPS 204)	Public key	—	1312	1312	bytes	NIST FIPS 204	Verified: ML-DSA-44
	Signature	—	2420	2420	bytes	NIST FIPS 204	Verified (FIPS 204)

Algorithm	Parameter	Current	Target	Theoretical	Unit	Source	Verification
	Signing speed	—	200	500	ops/s	Target impl.	HW-assisted target
	Verification	—	600	1000	ops/s	Target impl.	HW-assisted target
SLH-DSA-SHA2-128f	Signature	—	7 856	7 856	bytes	NIST FIPS 205	AF v2.0 (resources); auxiliary hash per NIPU v1.1, MEM v1.1
	Signing speed	—	15	30	ops/s	Target impl.	Target for future ASIC

> Withdrawn from v2.0: «SPHINCS+-256s, signature 29 792 bytes, “Verified: SLH-DSA SHA2-256s”» — an incorrect parameter set (canon: SHA2-128f) and an incorrect signature size (29 792 B belongs to 256s). The v2.0 glossary called SLH-DSA “stateful” — an error: FIPS 205 defines stateless hash-based signatures (the standard title appears in the v2.0 reference list [18]).

5. Hardware Resource Parameters (Layer I — instance table)

5.1 Layer A parameters (substrate-independent): policy types (5: Allow/Deny/Log/Quarantine/Defer); MAG target structure (§6); crypto module composition. — unchanged in substance.

5.2 Instance table (replaces the single-number table of v2.0, C3):

Parameter	RI-1 (VHDL, 2026-05)	RI-2 (SV, TCAD-2026-1011)	Evidence
HDL lines (SCC)	~340 (VHDL)	467 (SystemVerilog)	RI-1: AF v1.4; RI-2: TCAD manuscript
FPGA LUTs	3 000–5 000 (full implementation)	2 500–3 200 (TCB)	RI-1: MPS v2.0/AF v1.4; RI-2: TCAD; NIPU v1.1
FPGA flip-flops	4 000	—	RI-1: AF v1.4
BRAM	3	—	RI-1: PAR report
Clock frequency	100 MHz	142 MHz (post-synthesis)	RI-1: AF v1.4; RI-2: TCAD; NIPU v1.1
Device	Artix-7 100T (TQFP-144)	**XC7A35T**	RI-1: MPS v2.0; RI-2: AF v2.0 §9
Power envelope	~1.5 W (0.3 stat + 1.2 dyn)	< 1.3 W	RI-1: MPS v2.0; RI-2: AF v2.0
Static / dynamic power	~0.3 W / ~1.2 W	—	RI-1: measurement
Crypto module LUTs	~2 000	—	RI-1: AF v1.4

> Note: RI-2 cells without confirmed values are published as “—” and will be filled from working reports in a future revision.

6. Memory Architecture Parameters (MAG)

The v2.0 MAG table is unchanged. Addition (C5, D4):

The Memory Architecture Grid (MAG) defines the on-chip policy storage structure of the Amagi Framework. MAG uses OTP (One-Time Programmable) technology for immutable policy configuration.

Parameter	Symbol	Current	Target	Theoretical	Unit	Source	Verification
MAG dimensions	—	Not impl.	128x128	—	cells	MEM-SPEC v1.0	Architectural target
MAG total cells	—	—	16,384	—	cells	MEM-SPEC v1.0	128*128
Policy bits/cell	—	—	4	—	bits	MEM-SPEC v1.0	4-bit policy encoding
MAG total size	—	—	8,192	—	bytes	MEM-SPEC v1.0	16,384 x 4 bits = 8 KB
MAG lookup time	t_mag	—	<5	—	ns	MEM-SPEC v1.0	On-chip OTP, synchronous
Policy types	—	—	5	—	—	MEM-SPEC v1.0	Allow/Deny/Log/ Quarantine/Defer
OTP technology	—	—	eFuse/ antifuse	—	—	MEM-SPEC v1.0	eFuse for one-time config
OTP write endurance	—	1	1	—	cycles	Definition	One-time programming
Data retention at 85C	—	>20	>20	>100	years	Technology est.	Typical for eFuse >20 y
Radiation TID	—	—	—	>100	krad(Si)	Target	Space-grade antifuse OTP
Policy encoding	bits	4	4	4	bits	MEM-SPEC	Fixed architecture
MAG read current	I_mag_r	5	2	—	mA	Estimate	Reduced in ASIC
MAG standby current	I_mag_s	0.1	0.05	—	uA	Estimate	Ultra-low standby

Parameter	Value	Evidence
Policy update path	**Dual-Tier bank switching (ADR-002)**: active bank + staging bank; atomic activation; no rollback bypassing the SCC; corresponds to INV-004 (Controlled Mutability)	AF v2.0 §INV-004 (corrigendum v2.1 Planned); TM v1.1 SR-15

> This resolves the “OTP immutability of the MAG vs policy updateability (S10)” tension at the canon level: immutability applies to the active bank, mutability to the controlled bank switching. R1 Major 3 (dual-bank policy) is closed architecturally.

7. Domain-Specific Parameters

Domain-specific parameters define the safety requirements and operational constraints for each target industry vertical. All values are derived from applicable standards and regulatory frameworks.

7.1 Medical CPS (IEC 60601-1 / IEC 60601-2-24)

Parameter	Value	Unit	Source
Phase 1 threshold	Low-energy (OTP)	—	IEC 60601-1
Phase 2 threshold	High-energy (OTP)	—	IEC 60601-1
Inter-phase delay	>=200	ms	IEC 60601-2-24
Watchdog timeout	<=5	s	IEC 60601-2-24
Regulatory	IEC 60601-1, ISO 14971, FDA SaMD	—	IEC 62304

7.2 Automotive CPS (ISO 26262)

Parameter	Value	Unit	Source
Brake pressure	>=8	MPa	ISO 26262-3
Brake validation time	>=200	ms	ISO 26262-3
Throttle threshold	>20	% pedal	ISO 26262-3
Watchdog timeout	<=500	ms	ISO 26262 ASIL-D
Regulatory	ISO 26262, ECE/UNECE R79	—	ISO 26262-1

7.3 Aerospace CPS (DO-178C / EASA CS-25)

Parameter	Value	Unit	Source
Tank pressure	>=3.5	MPa	DO-178C
Valve position	Hardware-conf.	—	DO-178C
Inter-phase delay	>=500	ms	EASA CS-25
Igniter voltage	>2	kV	EASA CS-25
Watchdog timeout	<=2	s	EASA CS-25
Regulatory	DO-178C, EASA CS-25	—	DO-178C

7.4 Industrial Process Safety (IEC 61511 / IEC 62443)

Parameter	Value	Unit	Verification
Safety integrity level	SIL 2 (target 3)	—	Confirmed per IEC 61511-1
Logic solver reaction time	<=100	ms	Per IEC 61511-1 typical SIF
Sensor-to-actuator total time	<=500	ms	NAMUR NE 130 recommendation
SIF proof test interval	8760	h	Standard annual
Cybersecurity SL	SL 2 (target 3)	—	IEC 62443-3-3 zone/conduit
Max ambient temperature	-40 to +70	degC	Industrial range
Isolation voltage	1500	V DC	Galvanic isolation

7.5 Railway Signalling (EN 50126/8/9)

Parameter	Value	Unit	Verification
Safety integrity level	SIL 4 (target)	—	EN 50129
Tolerable hazard rate	1E-9 per hour	—	EN 50126
System reaction time	<50	ms	EN 50129 typical
Voltage transient	2 kV	—	EN 50124
EMC immunity	EN 50121-3-2	—	Standard
Temperature range	-40 to +85	degC	EN 50155
Vibration/shock	EN 61373 cat. 1	—	EN 61373

7.6 Nuclear I&C (IEC 61513, IEC 60880)

Parameter	Value	Unit	Verification
Safety category	Cat. A (target)	—	IEC 61226
Reaction time	<=300	ms	IEC 61513
Seismic qualification	0.3g PGA SSE	—	IEEE 344
Environmental EM/RFI	MIL-STD-461	—	System spec

Parameter	Value	Unit	Verification
Digital I&C diversity	Required	—	IEC 61513
Proof test interval	24 months	—	IEC 61513

7.7 Military / Defence (MIL-STD-882E, MIL-STD-461)

Parameter	Value	Unit	Verification
Mishap severity	Critical/Catastrophic	—	MIL-STD-882E
Probability target	1E-7 per flight hour	—	MIL-STD-882E
Reaction time	<200	ms	System spec
Environmental temp	-55 to +125	degC	MIL-STD-810
Vibration	Random 20g	—	MIL-STD-810
EMI/EMC	MIL-STD-461	—	Standard
TEMPEST	SDIP-27 Level A	—	Optional

8. Audit Trail Parameters

Audit trail parameters define the logging, storage, and verification mechanisms for immutable event recording within the Amagi Framework.

Parameter	Symbol	Current	Target	Theoretical	Unit	Source	Verification
Hash algorithm	—	SHA-256	SHA-256	SHA-3	—	NIST FIPS 202	SHA-256 per FIPS 180-4
Chain type	—	HMAC chain	HMAC chain	Merkle tree	—	NIST SP 800-185	HMAC chain provides tamper evidence
Key source	—	SRAM-PUF	SRAM-PUF	Quantum RNG	—	NIST SP 800-90B	Verified entropy source
Min entropy	H_min	>=256	>=256	1024	bits	NIST SP 800-90B	Confirmed by PUF testing
Signature upgrade	—	Dilithium	Dilithium	SPHINCS+	—	NIST FIPS 205	Planned upgrade path
Storage	—	Append-only	Append-only	Distributed	—	AF-COMP v1.4	Immutable log structure
Access	—	Read-only	Read-only	Immutable	—	AF-COMP v1.4	Architecture enforced
Max events stored	—	1E6	1E8	infinity	events	Design	Scalable based on external memory
Write latency	—	<1	<0.1	—	ms	Estimate	Fast non-volatile write
Block verification time	—	100	20	1	us	Est.	HMAC verification pipeline
Tamper evidence	—	Cryptographic	—	Immutable	—	Design	Cryptographic binding
Retention period	—	Indefinite	Indefinite	Indefinite	—	Policy	Lifetime of device
Export format	—	JSON log	JSON/CSV	Standard	—	Toolchain	Standard machine-readable

9. Comparison Matrix (Amagi vs Competitors)

The v2.0 comparison table with two corrections (C10):

- row «SIL/ASIL»: Amagi (Current) → **“SIL 2 @1oo1 / ASIL-B (mapping)”**; Amagi (Target) → **“SIL 3 @1oo2 / ASIL-D (mapping), Phase 2”**;
- row «Formal verification»: Amagi (Current) → **“SPIN (RI-1, RI-2)”**; Amagi (Target) unchanged (Isabelle/HOL roadmap).

10. Known Ambiguities and Resolutions (extended)

Ambiguity	Affected documents	Resolution in this sheet
TTP latency: 18–42 µs vs 1–10 µs	AF-COMP, MEM, NIPU	18–42 µs = total cutoff (RI-1, full); 1–10 µs = policy verification only
LUT 3 000–5 000 vs 2 500–3 200	AF v2.0, NIPU v1.1, TCAD	Different perimeters: full implementation (RI-1) vs TCB-only (RI-2); §5.2
SCC HDL: “VHDL ~340” vs “SV 467”	MPS v2.0 vs NIPU v1.1/ TCAD	Two instances: RI-1 (VHDL), RI-2 (SV, current reference); §1.2
SPIN states: 2.43E6 vs 25 927 680	MPS v2.0 vs TCAD	RI-1 vs RI-2 model runs; §18
Clock: 100 MHz vs 142 MHz	MPS v2.0 vs NIPU v1.1/ TCAD	RI-1 vs RI-2 post-synthesis; §5.2
LTL count: 16 (12+4) vs 15	MPS v2.0 vs TCAD/site	RI-1 (12 safety + 4 liveness) vs RI-2 (15; split per Open Spec v1.1); §18
SIL-2 vs SIL-3	AF-COMP, AOS, HARA, FMEDA	Formula §3.1: SIL 2 @1oo1 current, SIL 3 @1oo2 Phase 2
ASIL-B vs ASIL-D	AF-COMP, AOS, NIST-PROF	Domain mapping only; §3.4
PFDavg 3.6E-4 vs 6.9E-5; SFF 92% vs 93%; Type A vs B	MPS v2.0 vs FMEDA v1.0	FMEDA v1.1 is the sole normative source; §3.2
“TI 1 year” vs “10 years”	FMEDA/MPS vs AF v2.0	T1 ≠ T_M; §3.3
HMAC-SHA256 vs Kyber	AF-COMP, MEM, NIPU	HMAC = current; Kyber = target
MAG size not specified	AF-COMP, MEM, AOS	128×128 OTP = target (16 384 cells)
Watchdog 30 s universal	AF-COMP v1.3	30 s default; domain-specific in OTP
OTP immutability of the MAG vs policy updateability (S10)	MEM v1.1, AOS, public site	Dual-Tier bank switching (ADR-002); §6
Expansions of MAG: Memory Access Graph / Memory Architecture Grid / Mutable Authenticated Graph / Mutable Action Gate	Open Spec, Charter, NIST (canon) vs MPS vs FMEDA vs HARA	Canon: **Memory Access Graph** (MAG); MPS glossary corrected (§20)
Expansions of ATL: Access Trigger Layer / Audit Trail Log / Access Time Latency / Actuation Time Lock	Open Spec, MEM, AF v2.0 (canon) vs HARA vs MPS vs FMEDA	Canon: **Access Trigger Layer** (TTP engine + audit diode, MEM v1.1 §3.3)
SLH-DSA-SHA2-128f vs «SPHINCS+-256s»	NIPU/MEM v1.1, AF v2.0 vs MPS v2.0 §4.1	Canon: SHA2-128f, signature ~7 856 B; §4.1
Artix-7 100T vs XC7A35T; 1.5 W vs <1.3 W	MPS v2.0 vs AF v2.0	Per instance: RI-1 / RI-2 (§5.2)

11. Change Log

Version	Date	Author	Changes
0.1	2026-05-12	A.M. Burlai	Initial consolidation
1.0	2026-05-12	A.M. Burlai	Reviewed, harmonized
2.0	2026-05-13	AI-assisted extension	Extended parameters
2.0-V	2026-05-13	AI-assisted verification	Verification column
2.1	**2026-10-06**	A.M. Burlai (consistency freeze)	Two-layer model; RI-1/RI-2 register; §3 rebased on FMEDA v1.1; T1/T_M resolution; dual-bank (ADR-002); isolation profiles S/D; terminology fixes (TTP, Type B)

12. References

The v2.0 references [1]–[49] are retained with the following replacements and additions:

- [2] AOS v1.0 → Amagi Open Specification v1.1, DOI: 10.5281/zenodo.20753403;
- add: Burlai, A.M. (2026). AMAGI HARA v1.1, DOI: 10.5281/zenodo.20753408 (new version); AMAGI FMEDA v1.1, DOI: 10.5281/zenodo.20753410 (new version); AMAGI Traceability Matrix v1.1, DOI: 10.5281/zenodo.20753412 (new version); IAA v1.1, DOI: 10.5281/zenodo.20145198; AMAGI Master Glossary v1.0 (new record — DOI upon publication);
- RI-2 evidence: numeric values of the RI-2 column are anchored to the public artifact DOI (Wave 3: verification package, per VC v1.1 §9) upon its publication; until then — “per AMAGI TCAD manuscript TCAD-2026-1011 (rejected 2026-09-22; new submission in preparation)”,

The v2.0 reference list [1]–[49] follows, with the replacements above applied.

The following references form the normative basis for all parameters in this document.

[1] Burlai, A.M. (2025). Amagi Framework v1.4. DOI: 10.5281/zenodo.17849595. [2] Burlai, A.M. (2025). Amagi Open Specification v1.0. DOI: 10.5281/zenodo.17861024. [3] Burlai, A.M. (2025). Amagi-MEM v1.0. DOI: 10.5281/zenodo.17679327. [4] Burlai, A.M. (2025). Amagi-NIPU v1.0. DOI: 10.5281/zenodo.17682668. [5] Burlai, A.M. (2026). IAA v1.1. DOI: 10.5281/zenodo.20145198. [6] Burlai, A.M. (2026). AGIAM Charter v1.1. DOI: 10.5281/zenodo.20145094. [7] Burlai, A.M. (2025). Trade Secrets Notice. DOI: 10.5281/zenodo.17886699. [8] IEC 61508-1:2010. Functional safety of electrical/electronic/programmable electronic safety-related systems. [9] IEC 61508-2:2010. Requirements for electrical/electronic/programmable electronic safety-related systems. [10] IEC 61508-6:2010. Guidelines on the application of IEC 61508-2 and IEC 61508-3. [11] ISO 26262-1:2018. Road vehicles — Functional safety. [12] MIL-HDBK-217F. Reliability Prediction of Electronic Equipment. [13] NIST FIPS 180-4. Secure Hash Standard (SHS). [14] NIST FIPS 198-1. The Keyed-Hash Message Authentication Code (HMAC). [15] NIST FIPS 202. SHA-3 Standard: Permutation-Based Hash and Extendable-Output Functions. [16] NIST FIPS 203. Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM / Kyber). [17] NIST FIPS 204. Module-Lattice-Based Digital Signature Standard (ML-DSA / Dilithium). [18] NIST FIPS 205. Stateless Hash-Based Digital Signature Standard (SLH-DSA / SPHINCS+). [19] NIST SP 800-38D. Recommendation for Block Cipher Modes of Operation: GCM. [20] NIST SP 800-90B. Recommendation for the Entropy Sources Used for Random Bit Generation. [21] NIST SP 800-107. Revision 1: Recommendations for Applications Using Approved Hash Algorithms. [22] NIST SP 800-185. SHA-3 Derived Functions: cSHAKE, KMAC, TupleHash, and ParallelHash. [23] ISO 14971:2019. Medical devices — Application of risk management to medical devices. [24] IEC 60601-1:2020. Medical electrical equipment — Part 1: General requirements for basic safety. [25] IEC 60601-2-24. Particular requirements for the basic safety and essential performance of infusion pumps and controllers. [26] DO-178C. Software Considerations in Airborne Systems and Equipment Certification. [27] EASA CS-25. Certification Specifications and Acceptable Means of

Compliance for Large Aeroplanes. [28] IEC 61511-1. Functional safety — Safety instrumented systems for the process industry sector. [29] IEC 62443-3-3. Industrial communication networks — Network and system security — Part 3-3: System security requirements and security levels. [30] EN 50126. Railway applications — The specification and demonstration of Reliability, Availability, Maintainability and Safety (RAMS). [31] EN 50129. Railway applications — Communication, signalling and processing systems — Safety related electronic systems for signalling. [32] EN 50155. Railway applications — Electronic equipment used on rolling stock. [33] EN 61373. Railway applications — Rolling stock equipment — Shock and vibration tests. [34] IEC 61513. Nuclear power plants — Instrumentation and control important to safety — General requirements for systems. [35] IEC 60880. Nuclear power plants — Instrumentation and control systems important to safety — Software aspects for computer-based systems. [36] IEC 61226. Nuclear power plants — Instrumentation and control important to safety — Classification of instrumentation and control functions. [37] IEEE 344. IEEE Standard for Seismic Qualification of Equipment for Nuclear Power Generating Stations. [38] MIL-STD-882E. Department of Defense Standard Practice: System Safety. [39] MIL-STD-461. Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment. [40] MIL-STD-810. Environmental Engineering Considerations and Laboratory Tests. [41] Holzmann, G.J. (1997). The Model Checker SPIN. IEEE Trans. Software Engineering, 23(5), 279-295. [42] Klein, G. et al. (2009). seL4: Formal Verification of an OS Kernel. ACM SOSP. [43] Watson, R.N. et al. (2019). CHERI: A Hybrid Capability-System Architecture. IEEE S&P. [44] IEC 60068. Environmental testing — Part 2: Tests. [45] IEC 60529. Degrees of protection provided by enclosures (IP Code). [46] IEC 61000-4-3. Testing and measurement techniques — Radiated, radio-frequency, electromagnetic field immunity test. [47] CISPR 11. Industrial, scientific and medical equipment — Radio-frequency disturbance characteristics — Limits and methods of measurement. [48] JEDEC JS-001. Human Body Model (HBM) for Electrostatic Discharge (ESD) robustness. [49] DO-160G. Environmental Conditions and Test Procedures for Airborne Equipment.

13. Power, Thermal, Mechanical / 14. Environmental / 15. Certification Roadmap / 16. Communication / 17. Software & Firmware

This section consolidates power consumption, thermal operating ranges, and mechanical specifications for the Amagi Framework hardware implementation.

Key parameters are included in Section 5 (Hardware Resource Parameters). Additional details:

- Total FPGA power budget: ~1.5W (static + dynamic) for Artix-7 100T prototype
- ASIC target: <0.5W total power consumption at 28nm process node
- Thermal management: passive cooling sufficient for industrial temperature range
- Mechanical form factor: TQFP-144 (prototype) migrating to BGA-256 (ASIC target)

Environmental qualification parameters define the operating conditions and stress test requirements for the Amagi Framework across all target deployment environments.

Parameter	Symbol	Value	Unit	Source	Verification
Operating temperature	T_op	-40 to +85 (industrial)	degC	IEC 60068	Standard industrial range
Storage temperature	T_st	-55 to +125	degC	IEC 60068	Extended temperature
Relative humidity	RH	5-95 non-condensing	%	IEC 60068	Confirmed per 60068-2-78
Random vibration (op)	V_op	10-2000 Hz, 3.12 g RMS	—	MIL-STD-810	MIL-STD-810H method 514.6
Mechanical shock	Shock	50 g, 11 ms half-sine	—	MIL-STD-810	Method 516.6
ESD (HBM)	ESD_HBM	+/-2 kV (target +/-4 kV)	kV	JEDEC JS-001	JS-001 for HBM

Parameter	Symbol	Value	Unit	Source	Verification
Radiated immunity	—	80 MHz-1 GHz, 10 V/m	—	IEC 61000-4-3	Standard test level
Conducted emission	—	Class B per CISPR 11	—	CISPR 11	EMC compliance
Ingress protection	IP	IP54 (enclosure)	—	IEC 60529	Dust and splash proof
Salt fog (optional)	—	96 hours	—	MIL-STD-810	Method 509.5
Altitude	—	Up to 15,000	m	DO-160G	Section 4.6.1

The Amagi Framework supports standard industrial communication interfaces for integration with host systems and safety networks.

Supported Interfaces

- JTAG/IEEE 1149.1: Boundary scan and debug access
- SPI: Configuration and policy upload (master/slave mode)
- I2C: Sensor and actuator bus interface
- UART: Diagnostic logging and firmware update
- CAN/CAN-FD: Automotive and industrial fieldbus
- Ethernet 10/100: Network integration (optional)

Security Features

- All external interfaces implement rate limiting and watchdog monitoring
- Policy upload requires HMAC-SHA256 authenticated commands
- Debug interface disabled in production mode via OTP fuse

The Amagi firmware implements the safety logic, cryptographic operations, and communication protocols. Key platform parameters:

Firmware Architecture

- Core: Bare-metal C (no OS dependency for safety-critical functions)
- Code size: ~15 KB (safety core), ~45 KB (full firmware with crypto and comms)
- Stack usage: <2 KB maximum call depth
- Heap: Not used in safety-critical path (static allocation only)

Development and Build

- Compiler: GCC ARM/ RISC-V (verified toolchain)
- Build system: CMake with reproducible build verification
- Unit test coverage: 100% (safety-critical functions)
- Static analysis: MISRA-C:2012 compliant, Polyspace/ PC-lint validated

§15 edit (C9): Phase 1 is formulated as «IEC 61508-1/2: SIL 2 architectural alignment **@1oo1** achieved (FMEDA v1.1)»; Phase 2 (2027): «SIL 3 certification **@1oo2** (galvanic isolation, enhanced diagnostics)», «ISO 26262: ASIL-D route (domain mapping)». Addition: «Certification is performed per implementation instance by accredited bodies; architectural alignment ≠ certified product.»

18. Formal Verification and Testing Metrics (instance table)

Parameter	RI-1	RI-2	Evidence
Model / tool	SPIN v6.5 (Promela ~340 lines)	SPIN v6.5 (Promela, —)	§5.2
States explored	2.43E6	**25 927 680**	RI-1: verified run; RI-2: TCAD manuscript; **independently reproduced during TCAD peer review (Reviewer 2), statistics confirmed**

Parameter	RI-1	RI-2	Evidence
Transitions explored	8.71E6	—	RI-1: verified run
Depth	1500	—	—
LTL properties	16 (12 safety + 4 liveness)	**15 = SCC-S01…S09 (9 safety, PASS) + SCC-S10…S11 (architectural assumptions, trivially true by HW construction) + SCC-L01…L04 (4 liveness)**	RI-2: Open Spec v1.1 (formal verification); NIPU v1.1 §5. CEX L01–L04 are expected in the unconstrained-environment model; the L04 latch-up trap was eliminated by the FSM chain-reset refinement (NIPU v1.1 §5.1). **L04 status (updated 2026-10-25, T-002): the L04 counterexample is closed in the corrected model scc_amagi_v3 (watchdog inside SCC_Core + exclusive hw_emergency guard): L01–L04 verified, FA2 a theorem; Wave 3 package frozen.**
Assertion coverage (RTL)	>95% (QuestaSim)	—	RI-1
Fault injection / coverage	5000 faults / 99.2%	—	RI-1: fault sim on netlist
Unit test coverage (C)	100%	—	RI-1: gcov
HW/SW integration tests	156	—	RI-1

> **Rule (C8):** a value without an instance in the Evidence column is not published. Marketing use of «25.9M states» is permitted only with the «RI-2» attribute.

19. Lifecycle and Dependability Parameters

v2.0 unchanged in substance, with the C4 fix: «Design life / T_M: 10 years (87 600 h)»; «Proof test interval T1: 8 760 h (annual) — normative for every profile»; any wording that describes the 10-year figure as a proof-test interval is removed.

Lifecycle and dependability parameters define the operational lifetime, maintenance schedule, and reliability targets for the Amagi Framework.

Operational Lifecycle

- Design life: 10 years continuous operation (87,600 hours)
- Mission time (T_M): 87,600 hours per IEC 61508-6
- Proof test interval: 8,760 hours (annual)
- Firmware update: Supported via secure bootloader with HMAC verification

Maintenance and Support

- Mean Time Between Failures (MTBF): >100,000 hours (calculated)
- Mean Time To Repair (MTTR): <1 hour (field-replaceable module)
- Diagnostic test interval: 1 second (continuous online testing)
- Spare parts availability: 15 years minimum

20. Glossary of Selected Abbreviations

The v2.0 glossary table with the following corrections and additions:

- **TTP** — **Tautological Truth Predicate** (formal predicate enforced by the SCC; canonical per FMEDA v1.1 and Master Glossary v1.0). The obsolete «Temporal Triad Parameters» reading is removed; the temporal structure is named **Temporal Triad**;
- **ATL** — **Access Trigger Layer** (implements TTP delay at memory level, 0.5 ns–1 µs, and the audit transaction logger; canon per Open Spec v1.1, MEM v1.1, AF v2.0). Withdrawn: «Access Time Latency»;
- **MAG** — **Memory Access Graph** (128×128 eFuse/OTP policy matrix; canon per Open Spec v1.1, Charter, MEM v1.1). Withdrawn: «Memory Architecture Grid»;
- **MPE** — Memory Policy Enforcer; **SHMF** — Self-Healing Memory Fabric; **CRTM** — Core Root of Trust for Measurement (canon per Open Spec v1.1; «Code Root of Trust Mask» withdrawn per C13);
- **SLH-DSA** — State**less** Hash-Based Digital Signature (FIPS 205); canonical set: SLH-DSA-SHA2-128f.
- **T1** — Proof Test Interval (8 760 h);
- **T_M** — Mission / Design Life (87 600 h);
- **RI-1 / RI-2** — Reference Implementation instances (see §1.2);
- **PTCB** — Policy-Enforcing Trusted Computing Base;
- **Type A / Type B** — IEC 61508-2 subsystem classification (AMAGI TCB: **Type B**).

20.1 Glossary Table (v2.0, corrected)

Abbreviation Definition AF Amagi Framework AF-COMP Amagi Framework Component Specification AOS Amagi Open Specification ASIC Application-Specific Integrated Circuit ASIL Automotive Safety Integrity Level ATL Access Trigger Layer BRAM Block Random Access Memory CCF Common Cause Failure CPS Cyber-Physical System DC Diagnostic Coverage DTI Diagnostic Test Interval EASA European Union Aviation Safety Agency EMC Electromagnetic Compatibility EMI Electromagnetic Interference ESD Electrostatic Discharge FAA Federal Aviation Administration FIT Failure In Time (1E-9 failures/hour) FMEDA Failure Modes, Effects, and Diagnostics Analysis FPGA Field-Programmable Gate Array FSM Finite State Machine HBM Human Body Model HFT Hardware Fault Tolerance HMAC Keyed-Hash Message Authentication Code I&C Instrumentation and Control IEC International Electrotechnical Commission IP Ingress Protection / Intellectual Property ISO International Organization for Standardization LUT Look-Up Table MAG Memory Access Graph MEM-SPEC Amagi Memory Specification MIL-STD Military Standard MOSFET Metal-Oxide-Semiconductor Field-Effect Transistor NAMUR Normenarbeitsgemeinschaft fur Mess- und Regeltechnik NIPU Neural-Informed Processing Unit NIST National Institute of Standards and Technology OTP One-Time Programmable PFD Probability of Failure on Demand PFDavg Average Probability of Failure on Demand PGA Peak Ground Acceleration PLL Phase-Locked Loop PQC Post-Quantum Cryptography

Abbreviation Definition PUF Physically Unclonable Function RAMS Reliability, Availability, Maintainability, Safety RMS Root Mean Square RTL Register Transfer Level S-APL Strict Architectural Principle License SCC Safety Control Core SHA Secure Hash Algorithm SIL Safety Integrity Level SIF Safety Instrumented Function SL Security Level SPHINCS+ Stateless hash-based digital signature (SLH-DSA-SHA2-128f, FIPS 205) SPIN Simple Promela Interpreter (model checker) SRAM Static Random Access Memory TDF Time-Domain Firewall TID Total Ionizing Dose

TTP Tautological Truth Predicate (predicate); temporal structure: Temporal Triad VHDL VHSIC Hardware Description Language

End of Document This completes the Amagi Framework Master Parameter Sheet v2.0 Verified.

For the latest version, updates, and related documents, visit the AGIAM Technologies repository at DOI: 10.5281/zenodo.20145212

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